

ESTIMATION OF INTERNAL REPRESENTATION OF VOWELS IN COCHLEAR IMPLANT USERS

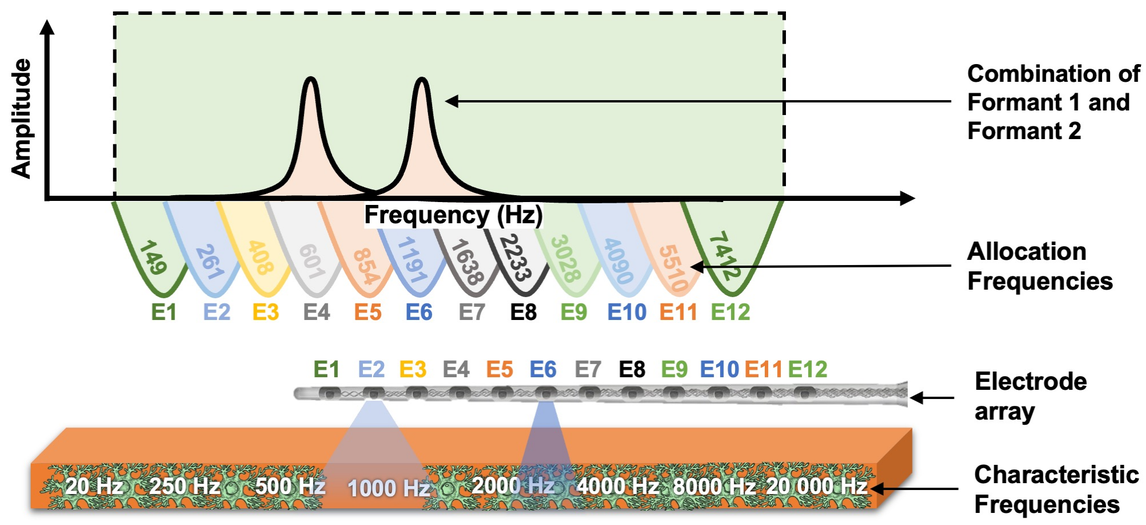
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The present study is part of a larger project aimed at providing clinicians with a testing solution to evaluate how cochlear-implant (CI) users adapt to their devices. Some CI users remain dissatisfied with their perception, especially in challenging listening situations. Here, we evaluate differences in the perception of a speech component - namely, vowels - in adults who were implanted post-lingually and have extensive experience with their device. In theory, their internal representations should not differ substantially from those of their normal-hearing peers; this is precisely what we investigate.

GOAL

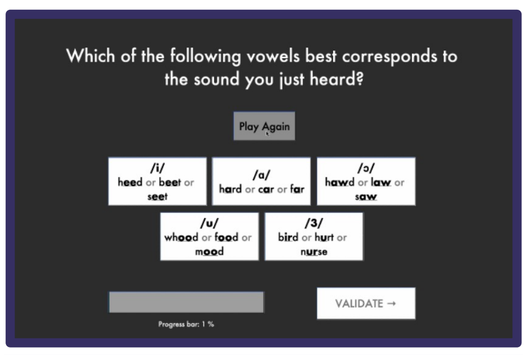
Does the internal representation of vowels differ between normal-hearing listeners and post-lingually implanted CI users with extensive implant experience?



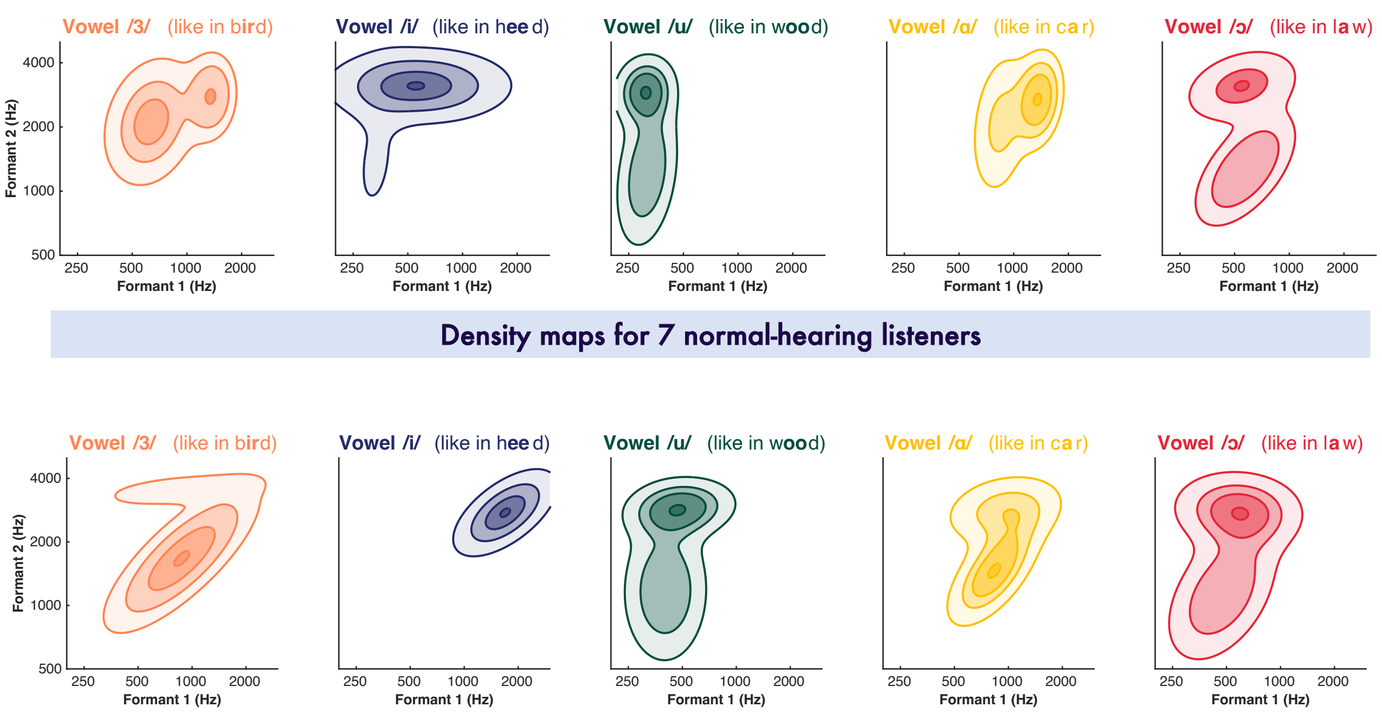
- INTERNAL REPRESENTATION OF VOWELS -

The experiment consisted of presenting several combinations of first-formant (**F1**) and second-formant (**F2**) frequencies. The specific F1 and F2 values were chosen according to each CI user's electrode mapping, so that each formant would stimulate a different electrode simultaneously. For each combination, participants selected the perceived vowel from a closed set (/ɑ/, /i/, /ɔ/, /u/, /ɜ/) and then rated their confidence on a Likert scale. From these two measures, individual perception heat maps for each vowel were computed for each CI user using a Bayesian model. The same procedure was carried out with seven normal-hearing listeners to generate reference vowel heat maps.

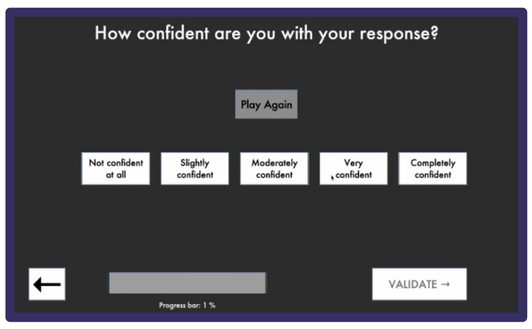
F1/F2 COMBINATIONS → **VOWEL IDENTIFICATION** → **CONFIDENCE SCALE** → **BAYESIAN ANALYSIS** → **DENSITY DISTRIBUTION**



Vowel identification

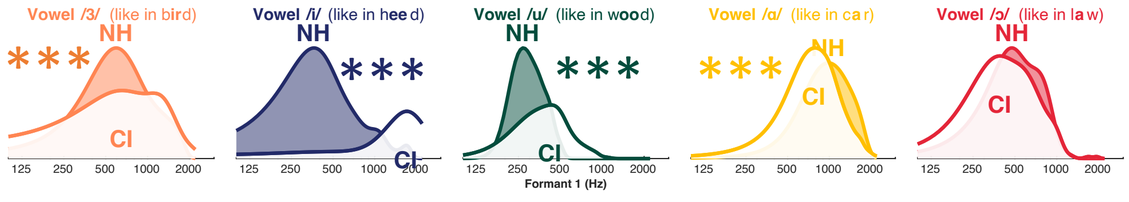


Density maps for 7 normal-hearing listeners

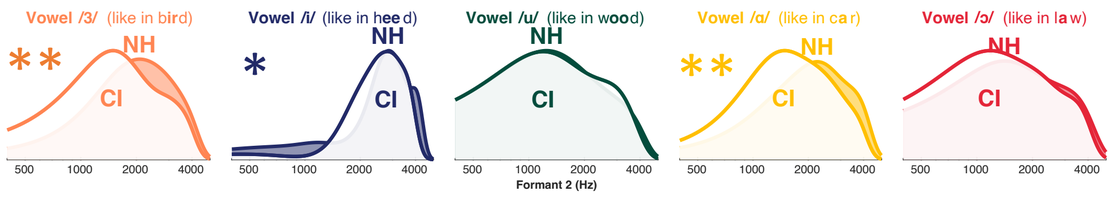


Confidence scale

Density maps for 8 cochlear implant users



F1 density maps, comparison between NH and CI



F2 density maps, comparison between NH and CI

PRELIMINARY CONCLUSIONS

ABOUT THE PARADIGM

- The interface is easy to use, intuitive, and offers reliable results.
- It is suitable for clinical use

ABOUT THE DIFFERENCE NH-CI

- The internal representation of vowels differs between NH and CI users
- CI users exhibit compression toward mid-frequencies compared to NH listeners.

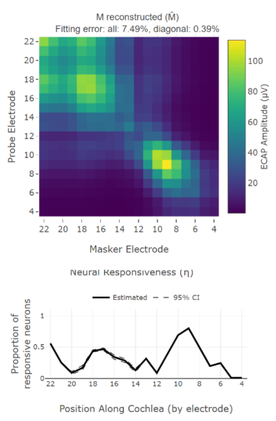
HOW TO EXPLAIN THESE DIFFERENCES?

INDIVIDUAL FACTORS

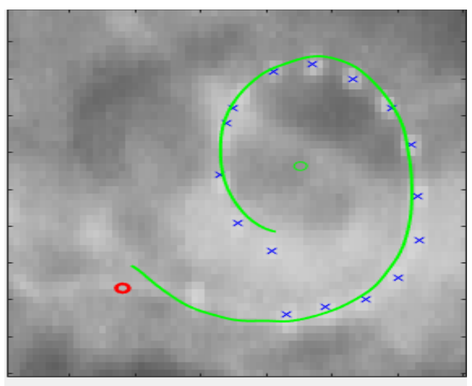
FREQUENCY-TO-PLACE FACTORS

- **FTPM[1]**
- **ENI[2]**

Estimation of the neural health (PECAP)



Estimate the degree of mismatch



NEXT STEPS

INCOMPLETE ADAPTATION?

Could the observed difference be due to incomplete adaptation related to individual factors?

- The electrode array position?
- The quality of the ENI?
- Interaction between these two factors?

[1]FTPM or Frequency-to-place mismatch it refers to the difference between the acoustic frequency information conveyed by a given electrode and the frequency corresponding to the tonotopic location stimulated by that electrode.

[2]ENI or Electrode Neuron Interface quality it refers to the effectiveness of transmission between a given stimulating electrode and the neuronal interface it stimulates.